SmartEAF™

Electric Arc Furnace Digital Twin & Process Optimizer — Product Datasheet

Extractmet Pvt Ltd

Reference plant: Industry-X

SmartEAF is a first-principles **static + dynamic** model of the electric arc furnace with an interactive operator console and decision-support layer. It predicts temperature, chemistry, slag, energy and heat loss across the whole heat, and is **recalibrated to modern 100–150 t UHP practice and India-relevant DRI charges**, validates against plant data, and guides operators to hit aim at lowest energy and highest yield — running fully offline in a browser or as a desktop application.

What it does

- **Static balance** — per-heat mass & energy balance: tap weight, yield, flux/oxygen/carbon demand, slag chemistry, specific energy.
- **Dynamic twin** — second-by-second trajectory of masses, temperatures, C/Si/Mn/P, slag, foaming, energy and losses.
- **Operator guidance** — colour-coded OK / WATCH / ACT diagnostics with concrete set-point recommendations.
- **Sensitivity studies** — one-factor sweeps, response surfaces and tornado rankings of the biggest levers.
- **Event log** — timed record of charges, additions, phase changes and endpoint.

Physics & chemistry modelled

- Elemental mass balances and heat balances at every time step.
- Reaction thermodynamics & kinetics: two-regime decarburisation (O_2-supply limited above critical C ~0.30 wt%, carbon-mass-transfer limited below; slag FeO coupled to bath carbon by the Turkdogan product (%FeO)(%C)=K(T)), Si/Mn/P oxidation, FeO formation & carbon reduction, CO post-combustion.
- Heat & mass transfer: arc-to-bath transfer, oxy-fuel burners, immersion melting, mass-transfer-limited refining.
- Scrap melting & **dissolution kinetics** (lime → slag basicity build-up).
- **Coupled conduction–convection–radiation** heat loss through the refractory wall (working / safety / insulation / shell) with shell & interface temperatures.
- Slag chemistry, basicity and a CO-driven foaming index modulating arc efficiency.

Validation (vs literature-typical bands)

Quantity	Model	Typical band
Specific electrical energy	~446 kWh/t	350–450
Total specific energy	~567 kWh/t	560–650
Tap-to-tap time	~52 min	40–60
Metallic yield	~97%	90–97
Slag basicity B2	~2.1	1.8–2.4
Oxygen use	~35 Nm³/t	20–40
Tap temperature	~1630°C	1620–1660

Reduced-order engineering coefficients are generic and are calibrated to the specific furnace before operational use.

Interfaces

- **Web console** — single self-contained HTML; interactive reactor schematic, all inputs with help, separate Static & Dynamic sections, event log, 8+ live plots. No install, no internet.
- **Desktop console** — native Python (Tkinter) operator desk with editable parameters, embedded plots, schematic, guidance, sensitivity and event log.
- **Library / API** — Python package for scripting, batch studies and integration.

Deployment & requirements

Web app	Any modern browser · fully offline
Desktop	Python 3.9+ (Windows / Linux / macOS)
Integration	On-prem link to historian / PLC / Level-2 (optional)
Data to calibrate	A handful of heat logs (charge, additions, power, tap analysis & temperature)

Packages

Assess (pilot)	Operate (annual licence)	Integrate (enterprise)
Model calibrated to 2–3 heats; baseline vs best-practice study; opportunity report; offline web app.	Everything in Assess; desktop + web consoles; sensitivity & scenario studies; quarterly re-calibration & support.	Everything in Operate; Level-2 / historian data link; live soft-sensing & advice; custom KPIs & on-site training.

Indicative packaging — final scope and pricing tailored per furnace and data availability.

Applications

Energy reduction (kWh/t) · Yield improvement · Productivity / tap-to-tap · Foaming & slag practice · DRI / hot-metal charges · Refractory heat-loss studies · Operator training · What-if & scenario analysis · Decarbonisation